



As in [Resource's Calculator](#), you can find that it can calculate accurate value of production with minimal errors. So this page will help you understand the math behind it.

City Production

Parameter

Resource type, Buildings, Population, Morale, City State and Hidden

Resource type

Each resource will have its own **base production**. These values will later be multiplied by others.

2100	1800	2100	1500	1200

Buildings

This only concerns "Resource Productions" part of the building's features. Arms Industry, Air Base and Naval bases have this feature. These '**modifier**' values are additive to each other then the sum is turned into '**multipliers**'.

Arms Industry

Level	1	2	5
Hit Points	10	15	30
Resource Production	+10%	+20%	+50%
Flat Resource Bonus	+100	+135	+200

Population

Population	1	2	3	4	5	6	7	8	9	10
Modifier	-80%	-60%	-40%	-20%	-	+5%	+10%	+15%	+20%	+25%

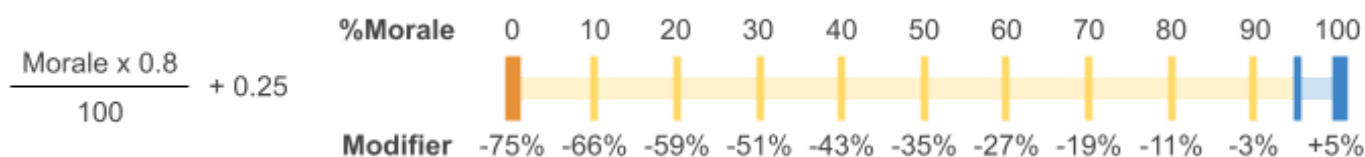
i Noted that Buildings and Population have intermediate value, In between the level or HP the "Resource Productions" is shifted linearly.

Example

- 7/10 HP Arms Industry = +7% instead of +10
- 5.5 Population = +2.5% instead of +5%

Morale

Will apply multiplier resulting from the following formula



City State

State	Homeland	Annexed	Occupied
Modifier	-	-50%	-75%

Hidden Factor

Some cities or in specific maps have these hidden multipliers.

Known Example

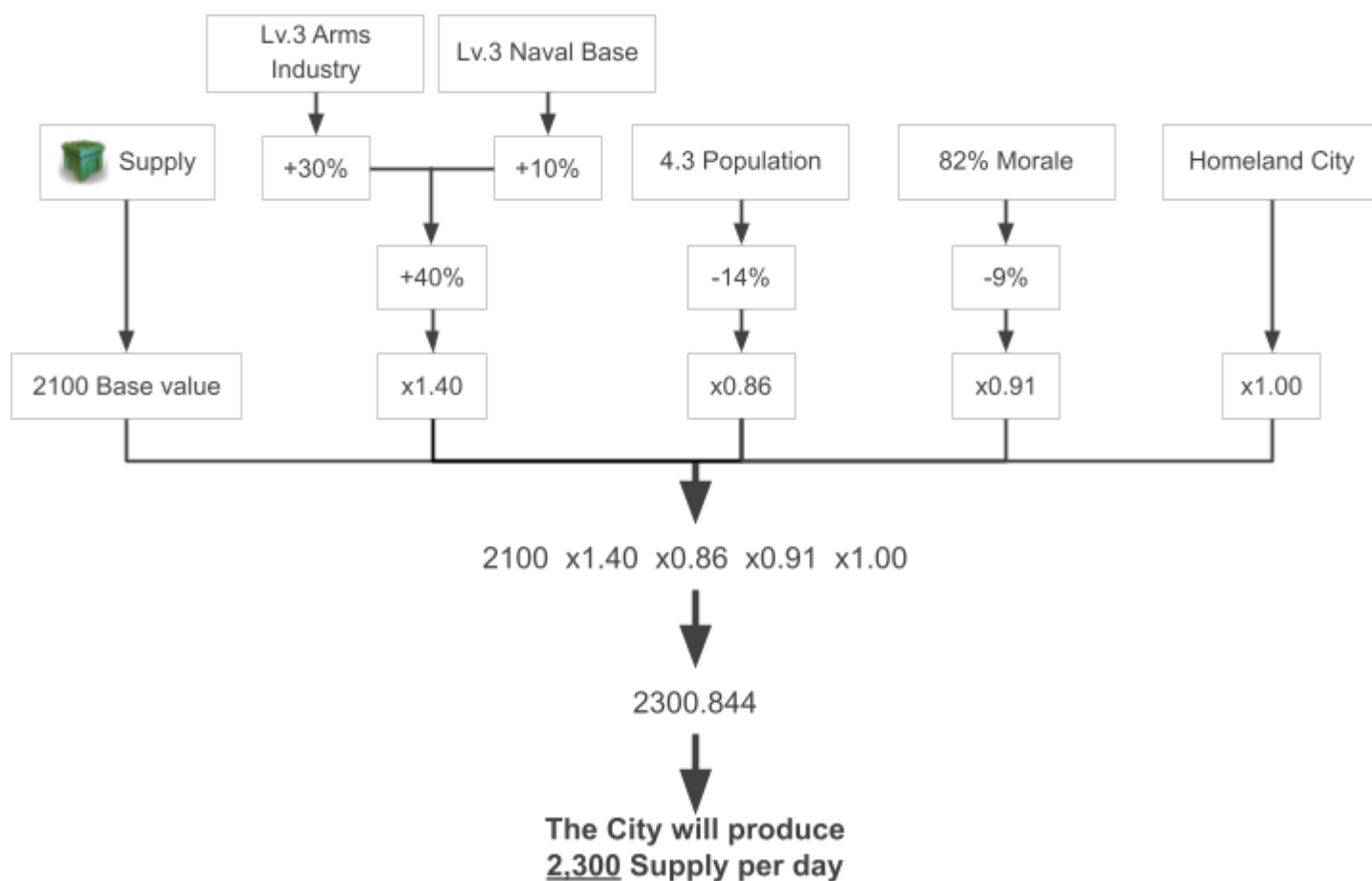
- 4x speed game, all city & province will be multiplied by '3/4'
- 10x speed game, all city & province will be multiplied by '1/2'
- Washington D.C. and Portland city of USA in WW3 will be multiplied by '2/3'
- Minsk city of Belarus in WW3 will be multiplied by '4/3'
- in USA Battleground, All Unplayable AI country will be multiplied by '2/3'
- All fuel city in Blood & Oil will be multiplied by '2/3'

Overview and Example

Now after all parameters are introduced, let's do a run on calculation!

Setting:

Supply Homeland city with 82% Morale, 4.3 Population, Lv3 Arms Industry, Lv3 Naval Port.



Simplified Production value

This value is mainly used in [Conflict of Nations' Map data](#) and somewhat arbitrary; due to different resources having their own base value, it is removed and the morale factor modified to be neutral at 70% morale in order to create a value of 1.00 for default setting of a city (empty city, morale 70%, 5 population) to be used for quick comparison between city and country overall production regardless of the resources.

Trivial

start resources: 20,000 supply, 15,000 component, 7500 fuel, 5500 electric, 5500 rare mat, 7500 man power, 75000 money

Edit log

- 8/2/2025 Updated Resource's Calculator link to a site, add simplified production value topic